PROJECT REPORT
ON

Inside Netflix's Big Data Brain: The Science of the Stream

Subject Name: Big Data Analytics
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Task 1: Big Data in Daily Life – Visual Storytelling

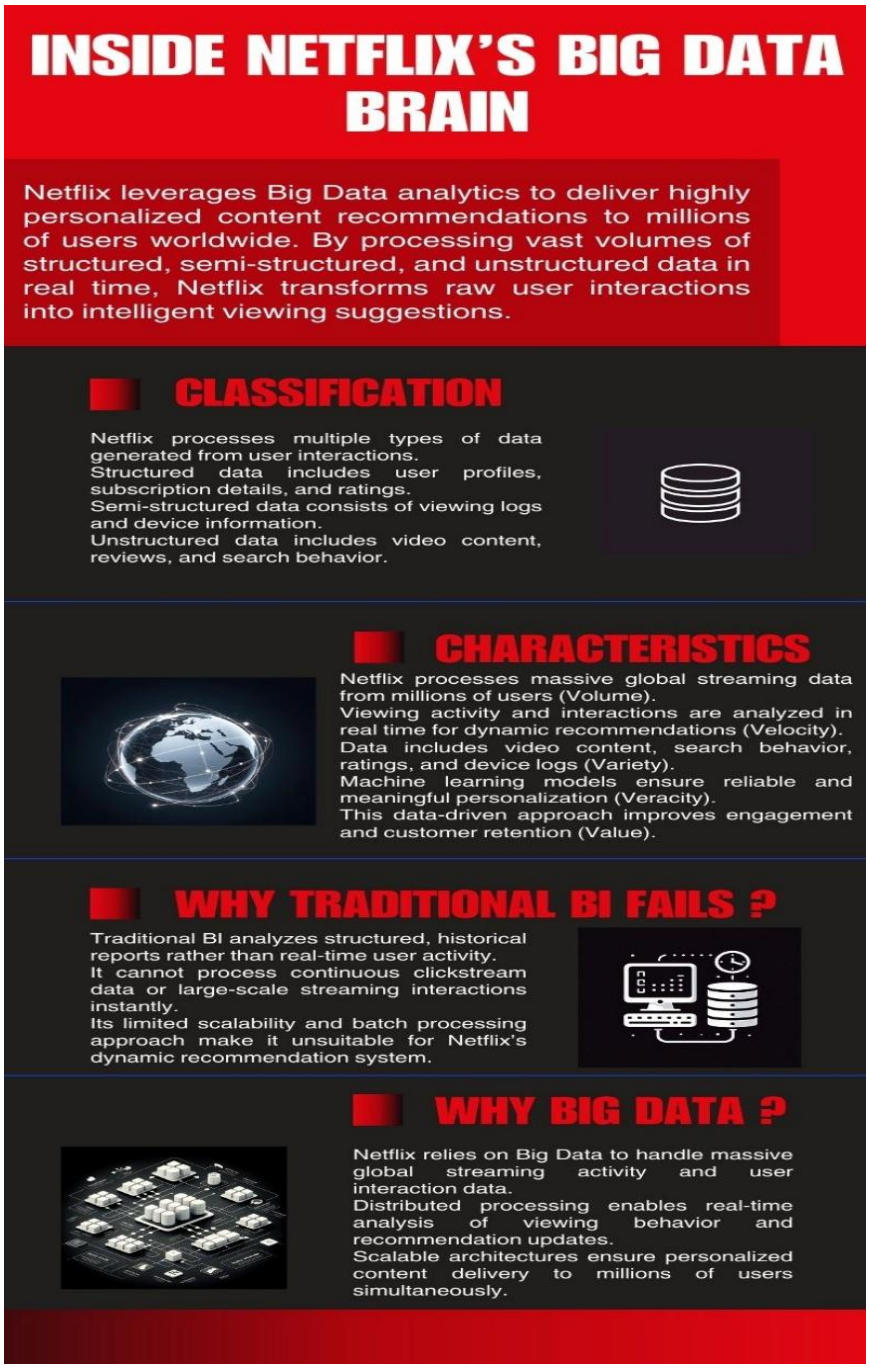

Fig 1.1: Inside Netflix's Big Data Brain

Task 2: BI vs Big Data - Role Play

Characters:

- John (Business Manager): Focused on user complaints and budget.
- Bob (Consultant): Focused on technical solutions and long-term strategy.

John: Hey Bob, our users are getting annoyed. Their "Recommended for You" lists are totally wrong lately. What's going on with our math?

Bob: The issue is that our old BI tools just can't keep up. Our SQL databases were made for small, neat data, not the crazy amount of streaming we do now.

John: But we've used this for years. Why is it breaking now?

Bob: It's how data has changed. We used to only track bills; now we track every single second a person is on a page. Our old systems aren't built for this messy, unstructured "Big Data".

John: Is that why the app crashes when everyone tries to watch Stranger Things at the same time?

Bob: Yep. That's a "Velocity" problem. Old BI can't handle millions of clicks every second without getting stuck.

John: And our server bills are huge! Why are we paying so much?

Bob: That's the "Volume" problem. We are saving way too many videos and logs. Trying to make old servers bigger costs 10 times more than moving to a Big Data setup.

John: You keep talking about "Hadoop". How does that save us money?

Bob: We need Hadoop because it uses "Distributed Storage". Instead of one super expensive computer, it spreads the data across a bunch of cheap ones. It's much easier to grow.

John: What about the messy stuff, like Twitter comments or phone data that doesn't fit in a table?

Bob: That's why we need "NoSQL". NoSQL doesn't need a perfect format. It lets us store all that "Variety" and analyse it right away without it breaking.

John: If we switch everything, what does Netflix actually get out of it?

Bob: Better "Personalization". By looking at the 5V's (Volume, Velocity, Variety, Veracity, and Value), we can guess what a user wants to watch before they even search for it.

John: So, using Hadoop and NoSQL stops "Churn", so people don't cancel their accounts?

Bob: Exactly. We keep our users happy and save about $1 Billion a year because people stay subscribed and we know exactly what shows to make next.

Task 3: Architecture Design Challenge

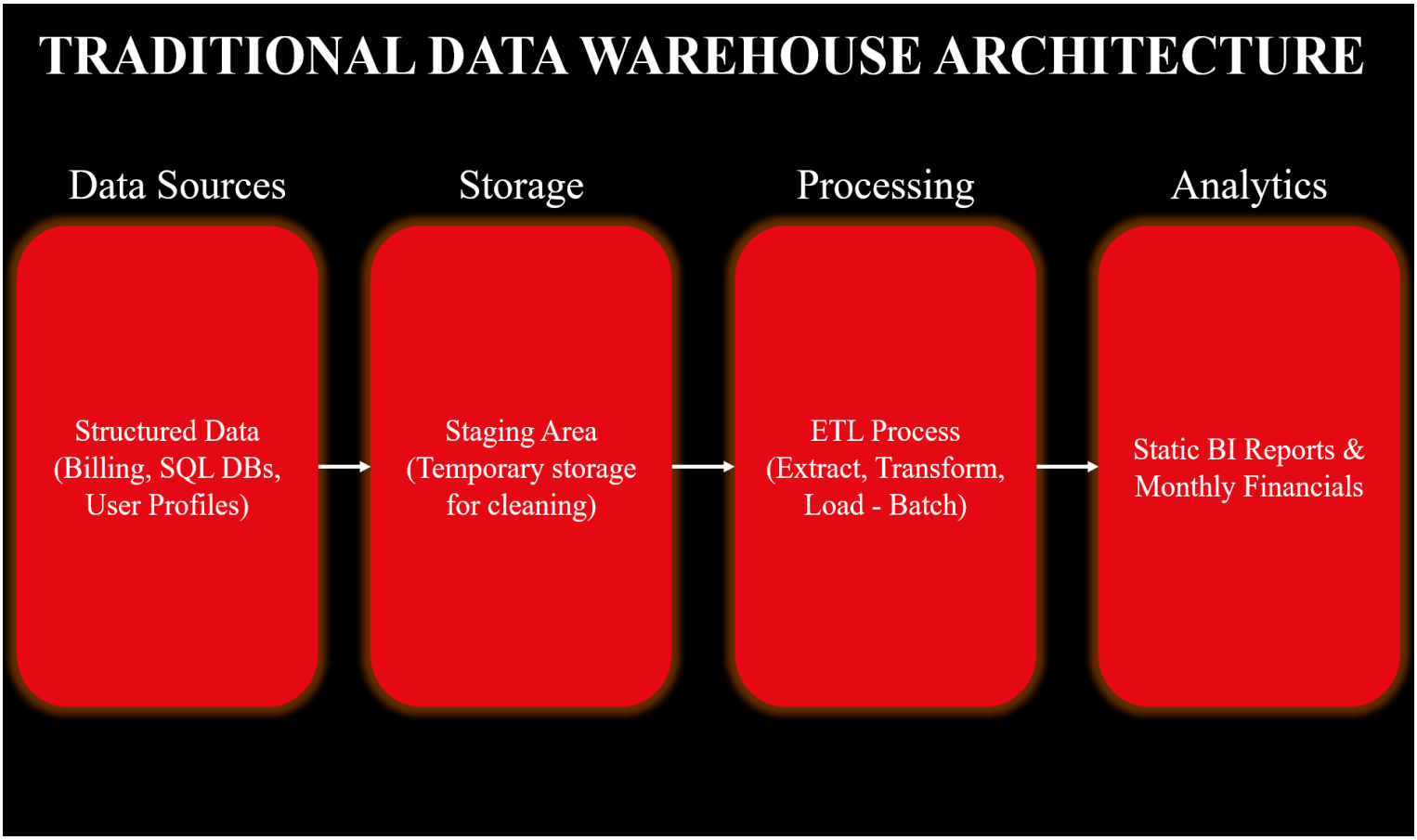

Fig 3.1: Traditional Data Warehouse Architecture

3.1 Traditional Data Warehouse Architecture:

This diagram shows how Netflix used to handle data before it became a global giant.

- Data Sources: In this old setup, we only focused on "Structured Data" like user profiles and billing information stored in SQL databases.

- Storage (Staging Area): Before the data could be used, it had to sit in a temporary "Staging Area" to be cleaned. This was a slow process that happened in batches.
- Processing (ETL): I used a process called ETL (Extract, Transform, Load). This worked like a filter, but it could only handle small amounts of data at a time.
- Analytics: The final result was just "Static Reports". This told managers what happened last month, but it couldn't give live movie recommendations to users.

3.1.1 Challenges and Limitations of the Traditional Setup:

While the Traditional Data Warehouse was the industry standard for a long time, it created several "bottlenecks" for a growing company like Netflix:

- Expensive Scaling (Vertical Scaling): To handle more data in this old system, we had to buy bigger, more expensive servers. This is called "Vertical Scaling," and it costs a lot more than using the "Horizontal Scaling" found in Big Data.
- The Batch Processing Delay: In this architecture, data is processed in "batches" (often once a week or once a day). This means managers couldn't see what was happening on the app in real-time, leading to outdated business decisions.
- Data Silos: Because this system only likes "Structured Data" (neat tables), all the "Unstructured Data" like user comments, social media feedback, and device logs were ignored and left in separate silos.
- Single Point of Failure: Traditional warehouses often rely on one main central server. If that server has a hardware issue, the whole analytics department stops working until it is fixed, whereas Hadoop is "Fault Tolerant" and keeps running.

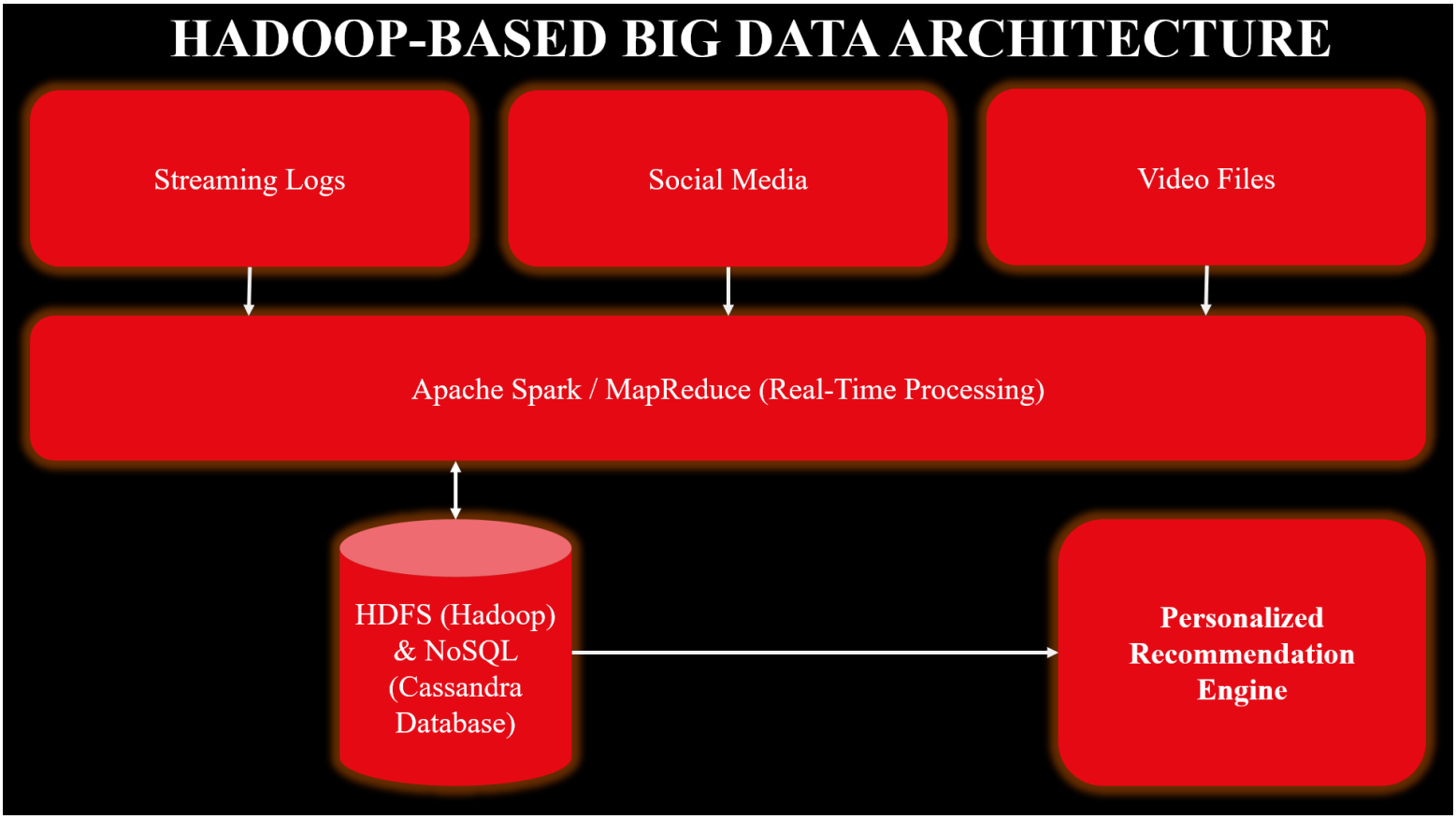

Fig 3.2: Hadoop-Based Big Data Architecture

3.2 Hadoop-Based Big Data Architecture:

This is the modern "Brain" of Netflix that we designed to handle the 5V's.

- Streaming Data Sources: Unlike the old way, this system takes in everything-live streaming logs, social media comments, and actual video files-all at once.
- Real-Time Processing (Apache Spark / MapReduce): This is the heart of the system. Instead of waiting for hours, Apache Spark processes millions of clicks every second while the user is still watching.

- Storage (HDFS & NoSQL): We use HDFS to spread data across many computers so it never runs out of space. We also use Cassandra (NoSQL) to store messy data like "likes" and "dislikes" without it crashing.
- Personalized Recommendation Engine: The final output is a smart engine that predicts exactly what you want to watch next, which keeps users happy and saves Netflix billions of dollars.

3.2.1 How the Hadoop Architecture Works:

The reason we switched to this design is that it doesn't just store data; it manages it across a "Cluster" of many computers. Here is the simple breakdown of what is happening inside that diagram:

- The "Divide and Conquer" Method (HDFS): When a massive video file or a million streaming logs come in, Hadoop doesn't try to save them on one disk. It breaks the data into small "blocks" and spreads them out. It also makes three copies of every block just in case one computer breaks (this is called "Fault Tolerance").
- The "Worker" System (MapReduce): In the old system, we had to move data to the computer to study it. In Hadoop, we do the opposite. We send the "work" to the data where it is already sitting. This saves a huge amount of time because we aren't moving giant files around the network.
- Master and Slaves: The architecture uses a "Master" node (the brain) that keeps track of where everything is, and many "Worker" nodes that do the actual heavy lifting.

Task 4: Analytics & Tool Match

Business Question	Analytics Type	Tool
What happened?	Descriptive Analytics	BI tools (Tableau, Power BI, or Netflix's internal dashboard)
Why did it happen?	Diagnostic Analytics	NoSQL / SQL Databases (To drill down into specific user logs and billing errors)
What will happen next?	Predictive Analytics	Hadoop / Spark (To process historical data and predict future viewing trends)
What action should be taken?	Prescriptive Analytics	Spark MLlib / AI Algorithms (The Recommendation Engine telling the user "Watch this next")

This table shows how we turn raw numbers into smart decisions. Each level of analytics helps Netflix answer a different kind of question to keep users happy.

- **Descriptive Analytics**: Simply looks at the past to see what movies were popular last week.
- **Diagnostic Analytics**: Investigates "why" something happened, like why a certain show is suddenly trending in India.
- **Predictive Analytics**: Uses historical data to guess what you will want to watch on a Friday night.
- **Prescriptive Analytics**: The most advanced step where the system automatically picks the "Top 10" list for your specific profile.

Bonus Challenge: "Explain Big Data to a 10-year-old"

Big Data Explained (The Minecraft Way)

What is it? Think about playing a simple game like "Flappy Bird". The game only has to remember your one score. That's regular data. Now, think about "Minecraft". There are millions of players building, breaking blocks, and chatting all at the same time. The world never ends! That massive, messy pile of info is Big Data.

The 3 Rules of the "Big World":

- **It's Huge (Volume):** The map is so big it won't fit on your laptop. It needs thousands of giant computers working together to hold all the blocks.
- **It's Fast (Velocity):** Every time someone clicks or moves, the game has to know right now. It's like a non-stop rain of information.
- **It's Everything (Variety):** It's not just blocks. It's the messages you type, the skins you wear, and even how long you look at a cow before hitting it.

Why does Netflix care? Netflix is like a "Pro Player" who has watched everyone play. Since it sees what millions of people like, it can tell you, "Hey, you liked that adventure map, so you'll probably love this dragon movie!" It uses all that big, messy data to find exactly what you want to watch next.

GitHub: https://github.com/Arpita-Achari/BDA-Assignment-1